

Label-Efficient Dark-Vessel Detection in SAR: A Controlled Comparison of Encoder Pretraining

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Abstract

Human-verified labels for vessels without matched Automatic Identification System reports are scarce. We test whether generic ImageNet, optical remote-sensing, or SAR pretraining reduces the labeled-data requirement for point detection in Sentinel-1 imagery. Two size-matched tracks use ViT-B/16 and ConvNeXt-V2-Base backbones. Four initialization roles within each track share the input, detector, schedule, scene subsets, seed, and strict-FP32 execution. The design contains 32 cells across four nested label budgets. All 32 cells completed training on H100 hardware, and this report presents the resulting four-fraction comparison from hash-bound evidence. Transfer helps most under label scarcity, but the sign of the SAR-versus-optical contrast depends on the architecture, and generic ImageNet initialization leads both tracks at full data. Reported values are corrected development-selection F1. The 50-scene human-verified set remains sealed.

1 Introduction

Sentinel-1 synthetic-aperture radar (SAR) supplies maritime imagery through clouds and without daylight [7]. Analysts can use it to find vessels that lack a contemporaneous matched Automatic Identification System report. These dark-vessel candidates matter for fisheries enforcement, maritime safety, and sanctions monitoring. They also present a difficult learning problem: a vessel occupies few pixels in a large scene, coastal infrastructure creates strong clutter, and human verification costs time. The xView3-SAR benchmark provides georeferenced point labels and a geographic matching protocol for this setting [3].

Transfer learning may reduce the number of target labels required for a useful detector. Generic ImageNet features offer broad visual structure. Optical remote-sensing checkpoints add overhead-scene statistics, while SAR checkpoints share the sensing modality. Public checkpoints confound source domain with model family and pretraining objective, so an uncontrolled comparison cannot isolate transfer effects.

We use two architecture-matched tracks. Four ViT-B/16 arms and four ConvNeXt-V2-Base arms differ only in initialization within their track: random, ImageNet, optical remote sensing, or SAR. Matched roles across tracks show whether a within-track pattern survives a change in architecture. We make no new detector or pretraining claim. The contribution is a controlled 32-cell design, one corrected evaluation contract, and a provenance-bound H100 result path.

2 Dataset and Evaluation Protocol

The frozen xView3 cohort contains 111 training scenes, 23 development scenes, and 16 test scenes. A separate set of 50 human-verified scenes remains sealed for one final evaluation. Scene identifiers, rather than chips, define every split. The four training budgets contain 12, 28, 56, and 111 scenes and satisfy $S_{10} \subset S_{25} \subset S_{50} \subset S_{100}$. All arms see the same scene set at a given budget.

The preprocessing path converts each dual-polarization GRD scene into a fixed three-channel tensor $[VH, VV, VH - VV]$ in decibels. It computes normalization statistics from training scenes only and creates overlapping 800-pixel chips. Optimization samples shared 512-pixel crops from these chips.

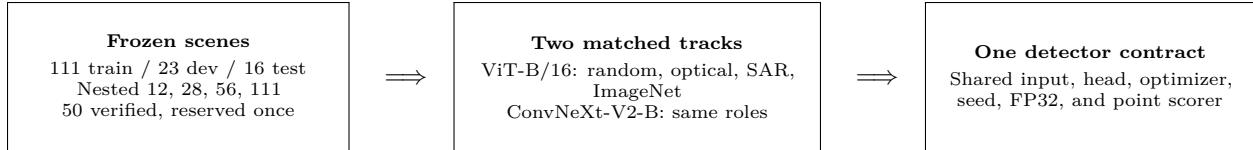


Figure 1: Controlled study design. Initialization is the only variable within each architecture track. Matched roles are compared descriptively across tracks; the final human-verified set remains sealed until the cohort and analysis code are frozen.

Table 1: Controlled initialization roles. Source heads are discarded.

Role	ViT-B/16	ConvNeXt-V2-Base
Random	fresh initialization	fresh initialization
Optical	SatDINO, fMoW-RGB	BigEarthNet Sentinel-2
SAR	SARMAE, SAR-1M	BigEarthNet Sentinel-1
ImageNet	supervised AugReg	FCMAE + supervised

The corrected label contract treats a HIGH- or MEDIUM-confidence row as a positive only when `is_vessel` is true. HIGH/MEDIUM non-vessels form background; LOW-confidence rows define ignore regions. The eight development scenes used during training contain 517 positives, 107 background non-vessels, and 118 ignores under this contract. Training never stages test or human-verified labels.

The frozen scorer greedily matches predictions and ground truth within 200 m, then aggregates precision, recall, and F1 across scenes. Each cell selects its threshold on development predictions and binds that operating point to the selected checkpoint. The committed evidence reports development-selection F1. It never substitutes that value for test F1, dark-vessel recall, or near-shore performance; held-out values render only from their own immutable results.

3 Related Work

xView3 frames dark-vessel detection as localization in Sentinel-1 SAR rather than box classification [3]. A CenterNet-style heatmap fits the point annotations and distance metric without inventing box extents [9]. We use this established detector form to hold the task head fixed.

The ViT track compares supervised ImageNet AugReg [4], optical SatDINO trained on fMoW-RGB [5], and SARMAE trained on SAR-1M [2]. SatDINO and SARMAE also differ in objective, so their gap is a released-checkpoint comparison rather than a pure modality intervention. The CNN track uses ConvNeXt-V2-Base [8]. Its optical and SAR arms come from matching BigEarthNet Sentinel-2 and Sentinel-1 land-cover models [6, 1]; this pair gives the cleaner modality contrast. ImageNet controls match the generic source dataset but not full training history across architectures.

4 Methods

Every arm predicts a stride-4 vessel heatmap from the same input. Gaussian targets mark positive points, and ignore neighborhoods mask LOW-confidence labels. The detector decodes local peaks above a permissive candidate floor, maps them to scene coordinates, and applies distance-based non-maximum suppression before scoring. ViT and CNN adapters reconcile native feature strides with the shared 256-channel head.

One PyTorch Lightning module and data module serve all cells. AdamW uses a 10^{-4} learning rate, 0.05 weight decay, layer-wise decay 0.65, five warm-up epochs, and cosine decay across at most 50 epochs. Full-scene development evaluation occurs every five epochs; early stopping waits four evaluation cycles. All runs use seed 0, micro-batch 16, accumulation 1, and end-to-end fine-tuning.

The corrected campaign runs one process per NVIDIA H100 in Lightning `32-true`. CUDA matrix-multiply TF32 and cuDNN TF32 stay disabled. Runtime receipts bind code, configuration, environment, GPU identity, checkpoint hashes, and completion markers. The evidence generator rejects a marker that does not hash to its cohort binding, a metric inconsistent with its own TP/FP/FN counts, a best-dev value that disagrees with the training curve, a mixed hardware class, or a partial held-out cohort.

Table 2: Development-selection F1 for all 32 cells. Columns give the label budget as a fraction of the 111 training scenes (scene counts in parentheses). Every cell shares the detector, schedule, seed, and strict-FP32 recipe; initialization is the only variable within a track.

Track	Initialization	10% (12)	25% (28)	50% (56)	100% (111)
ViT-B/16	Random	0.8030	0.8563	0.8912	0.8958
	Optical RS	0.8902	0.8802	0.9136	0.9268
	SAR	0.8857	0.8624	0.9161	0.9289
	ImageNet	0.8761	0.8716	0.9284	0.9399
ConvNeXt-V2-B	Random	0.7992	0.8167	0.8664	0.8919
	Optical RS	0.7992	0.8063	0.8586	0.8636
	SAR	0.8762	0.8868	0.8998	0.9068
	ImageNet	0.8856	0.9005	0.9221	0.9387

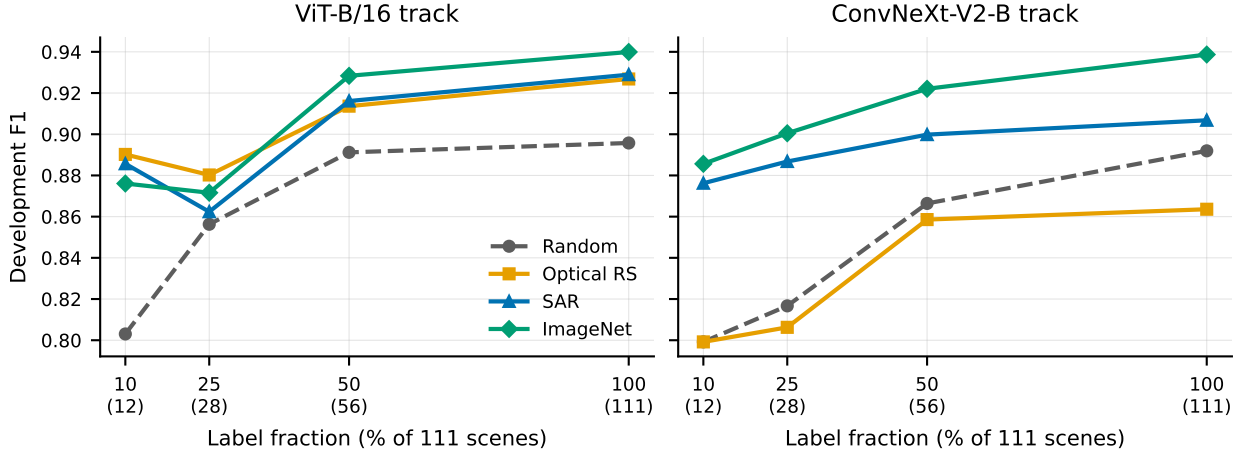


Figure 2: Development-selection F1 against the label budget. Colors identify initialization roles; the dashed gray curve is each track’s random floor. Both panels share the y-axis.

5 Results and Analysis

Every value in this section comes from the committed evidence tree. The fail-closed generator re-derives each number from the frozen training cohort, the hash-bound completion markers, and the full training curves; a single inconsistent byte stops the build. All 32 cells completed training on one NVIDIA H100 80GB HBM3 class in strict FP32 (548.4 GPU-hours, code `1a82d508`, cohort `9b1ba03e`, seed 0). Reported F1 is corrected development-selection performance: each cell selects its checkpoint and operating threshold on the eight development scenes. Table 2 lists the matrix and Fig. 2 shows the label-fraction curves.

Transfer helps under scarcity, with one exception. At the 12-scene budget, every pretrained ViT arm clears its random floor by a wide margin: optical adds +0.087, SAR +0.083, and ImageNet +0.073 F1 over 0.8030. In the CNN track, SAR adds +0.077 and ImageNet +0.086 over 0.7992. The exception is CNN-optical (BigEarthNet Sentinel-2), which only matches its floor at 10% (both 0.7992) and falls below it at 25% and 50%. The land-cover classification source appears to teach this backbone features that vessel detection must first unlearn.

The SAR-versus-optical contrast is architecture-dependent. The CNN track orders SAR above optical at every budget, and the gap is large under scarcity (0.8762 versus 0.7992 at 12 scenes). The ViT track disagrees below half data: optical leads SAR at 10% (0.8902 versus 0.8857) and at 25%, and SAR recovers only from 56 scenes upward. A modality-matched encoder is therefore not uniformly the best scarce-label choice; the architecture and pretraining objective moderate the effect. Because the released checkpoints also differ in objective and source volume, these contrasts describe available checkpoints rather than a causal effect of modality alone.

Generic ImageNet wins at full data. With all 111 scenes, ImageNet initialization leads both tracks (0.9399 and 0.9387). Domain pretraining matters most where labels are scarce; with enough target data, generic visual structure plus full fine-tuning closes the domain gap.

Label efficiency. The best 12-scene transfer arms nearly match the 111-scene random floors: ViT-optical reaches 0.8902 against the ViT floor’s 0.8958 at full data, and CNN-ImageNet reaches 0.8856 against 0.8919. Roughly ninety percent of the labeling budget buys what a suitable public checkpoint provides.

5.1 Held-out results

The 16-scene test split had not been scored under the held-out contract when this report was built. The contract freezes the cohort first, applies each cell’s development-bound threshold verbatim, and writes one immutable result per cell; the generator admits the test column only when all 32 results validate. Two cross-hardware control cells scored locally under this contract (RTX 5070 Ti, strict FP32) showed development-to-test changes of -0.080 and -0.069 , consistent with development-selection optimism rather than threshold-transfer failure.

The 50 human-verified scenes remain sealed. Their once-only evaluation is deferred until after this submission’s cohort and analysis code freeze.

6 Conclusion and Future Work

We built a controlled two-track experiment for measuring transfer under four nested label budgets. One detector, one corrected label contract, and one strict-FP32 H100 recipe constrain all 32 cells, and every published number is re-derived from hash-bound evidence at build time.

The completed grid supports three conclusions. First, a suitable public checkpoint recovers most of what roughly ninety percent more labels buy a randomly initialized detector. Second, matching the sensing modality is not uniformly the right scarce-label choice: SAR pretraining dominates in the CNN track at every budget, while the ViT track prefers the optical checkpoint until half the label budget. Third, generic ImageNet initialization wins both tracks at full data. These remain corrected development-selection results from one seed; they support no variance claim.

The next step scores the frozen 16-scene test split under the held-out contract, which applies each cell’s development-bound threshold verbatim. The 50 human-verified scenes then support one sealed evaluation of dark-vessel recall and near-shore performance for follow-on work. John Roth and Kyle Wagner jointly designed the study, implemented its data, model, training, and evaluation code, ran the experiments, analyzed the evidence, and wrote this report.

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A Full Evidence Inventory

Table 3 lists every cell with its development-selection F1, its checkpoint-bound operating threshold, the epoch of the selected checkpoint, and the held-out test F1 where admitted. All values are re-derived at build time from the committed cohort, markers, and training curves; dashes mean the fail-closed generator admitted no value, never zero performance.

Table 3: Per-cell evidence inventory (development-selection values; test column admitted only when all 32 immutable held-out results validate).

Cell	Budget	Dev F1	Threshold	Best epoch	Test F1
ViT random	10%	0.8030	0.822	19	—
ViT random	25%	0.8563	0.890	29	—
ViT random	50%	0.8912	0.853	44	—
ViT random	100%	0.8958	0.778	14	—
ViT optical	10%	0.8902	0.888	49	—
ViT optical	25%	0.8802	0.910	24	—
ViT optical	50%	0.9136	0.902	39	—
ViT optical	100%	0.9268	0.859	34	—
ViT SAR	10%	0.8857	0.587	4	—
ViT SAR	25%	0.8624	0.768	4	—
ViT SAR	50%	0.9161	0.855	39	—
ViT SAR	100%	0.9289	0.904	24	—
ViT ImageNet	10%	0.8761	0.906	39	—
ViT ImageNet	25%	0.8716	0.929	34	—
ViT ImageNet	50%	0.9284	0.844	49	—
ViT ImageNet	100%	0.9399	0.894	49	—
CNN random	10%	0.7992	0.716	29	—
CNN random	25%	0.8167	0.819	44	—
CNN random	50%	0.8664	0.857	44	—
CNN random	100%	0.8919	0.910	49	—
CNN optical	10%	0.7992	0.749	29	—
CNN optical	25%	0.8063	0.822	44	—
CNN optical	50%	0.8586	0.879	34	—
CNN optical	100%	0.8636	0.878	44	—
CNN SAR	10%	0.8762	0.776	34	—
CNN SAR	25%	0.8868	0.877	14	—
CNN SAR	50%	0.8998	0.893	24	—
CNN SAR	100%	0.9068	0.832	4	—
CNN ImageNet	10%	0.8856	0.869	4	—
CNN ImageNet	25%	0.9005	0.906	39	—
CNN ImageNet	50%	0.9221	0.899	19	—
CNN ImageNet	100%	0.9387	0.894	44	—

The campaign’s sanitized operator status snapshot (captured before the final cells closed) remains committed beside the evidence tree as a historical record of execution state at the deadline.

B Reproducibility and Audit Record

The evidence tree records the frozen training cohort, each cell’s hash-bound completion marker, its full training curve, and a sanitized runtime receipt (H100 hardware class, source SHA, strict-FP32 backend state). The generator validates marker-to-cohort hashes, metric consistency, and curve agreement before writing a manuscript artifact. Checkpoint bytes stay outside the repository; their SHA-256 bindings are published for archive verification. The submission excludes imagery, labels, weights, checkpoints, run trees, credentials, and private paths.

The detector uses 512-pixel crops, a stride-4 heatmap, Gaussian standard deviation two output pixels, a candidate floor of 0.05, and 120 m distance NMS. Training uses at most 50 epochs with development evaluation every five epochs. The project fixes seed 0 and reports point estimates without uncertainty bands.

C Qualitative-Evidence Boundary

The committed evidence contains metrics, curves, and receipts rather than exported H100 scene overlays. We omit qualitative examples instead of substituting engineering probes or opening sealed imagery. After the

held-out results land through the fail-closed path, we will sample successes and errors from the pre-registered held-out outputs.